

YUHAN WANG (王钰涵)

CONTACT

Department of Agricultural and Applied Economics
University of Wisconsin - Madison
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EDUCATION

University of Wisconsin - Madison

M.S., Agricultural and Applied Economics

2019 - 2021

M.S., Computer Science

2021 - 2023

Ph.D., Agricultural and Applied Economics (in progress)

2021 - Expected May 2027

Peking University

2015 - 2019

B.S., Life Sciences (Bioinformatics track)

B.A., Economics

Uppsala University

Summer 2017

Summer Session, Sweden Nature

RESEARCH INTERESTS

Agricultural Economics, Environmental and Resource Economics, Agricultural Finance, Empirical Industrial Organization, Applied Econometrics, Climate Adaptation

RESEARCH PROJECTS

Working Papers

- *Supply Networks and Market Power in U.S. Poultry Procurement*
- *A Text-Based Measure of Farm Bill Uncertainty and Its Effects on Agricultural Credit*. Under revision at *Journal of Agricultural Economics*.
- *Climate-Conditioned Comparative Advantage, Persistence, and U.S. Corn-Soy Land Allocation* (with Xiaodong Du and Xueying Sun). Under review at *Journal of Environmental Economics and Management*.

Selected Work in Progress

- Concentration, Bottlenecks, and Resilience in the U.S. Beef Supply Chain (with Xiaodong Du)
- How Does Climate Change Affect the Farmland Allocation and Trade Flow in the U.S.?
- U.S. Meat Market Demand Analysis using Nielsen Data
- Food Policy Uncertainty, Corporate R&D Investment, and Firm Productivity
- Media Bias and the Reporting of Economic News in the United States
- Market Structure and Price Dynamics in the Texas Electricity Market (initial ERCOT and EIA data cleaning)
- Weather Extremes and Tail Risk in U.S. Corn and Soybean Production

RESEARCH EXPERIENCE

Graduate Researcher, U.S. Meat Market Demand Analysis, UW-Madison AAE

2020 - Present

Individual Researcher, VIX Trading Strategies, City University of Hong Kong

2018

Researcher, Dimensionality Reduction Algorithms in Julia, Peking University

2018 - 2019

Summer Visiting Researcher, Uppsala University

2017

PROFESSIONAL EXPERIENCE

Quantitative Strategy Researcher, Alphafund Investment Company, Beijing

2018 - 2019

Audit Intern, HSYH Accounting Firm, Beijing

2018

TEACHING EXPERIENCE

AAE421: Economic Decision Analysis, FA22

TA

AAE722: Machine Learning in Applied Economic Analysis (001), GFF SU21

TA

AAE722: Machine Learning in Applied Economic Analysis (001), FA22

TA

Mathematics and Chemistry Teacher, TAL Education Group, Beijing

2017 - 2018

HONORS, SCHOLARSHIPS, AND FELLOWSHIPS

Deborah and David Penn Ag and Applied Economics Graduate Student Fund

2025

Barbara Forrest Student Award	2024
Deborah and David Penn Ag and Applied Economics Graduate Student Fund	2023
UW-Madison Graduate School Conference Presentation Award	2022
First Prize, National Mathematics Competition	2015
First Prize, National Chemistry Competition	2015

CONFERENCES & SEMINAR PRESENTATIONS

2025: AAEA & WAEA Joint Annual Meeting, Denver, CO, July 27-29; Poster Presentation: *Agricultural Lending Under Policy Uncertainty* (with Sheldon Du)

2024: AAEA Annual Meeting, New Orleans, LA, July 28-30; Poster Presentation: *How Does Climate Change Affect the Farmland Allocation and Trade Flow in the U.S.?*

SERVICES & OTHER ACTIVITIES

UW Madison AAE Seminar Organizer	2023 - 2024
Core Member, Peking University Life Science Industry Association	2015 - 2017
Leader, Peking University School of Life Sciences Student Union	2015 - 2017
Team Member, Peking University Yuanhuo Chess and Card Club	2019

SKILLS

Computing	C/C++, R, Python, Julia, SQL, Excel, MATLAB, Stata, Java, Microsoft Office, LaTeX
Languages	English (fluent), Mandarin Chinese (native)

ABSTRACTS

Supply Networks and Market Power in U.S. Poultry Procurement

Abstract: I construct the first shipment-based U.S. poultry procurement network, identifying the plants and firms competing for local purchases. Production and consumption maps alone do not reveal these supplier choice sets. A demand-and-supply model combines heterogeneous procurement decisions with joint pricing by multi-plant firms. Wholesalers are 3.33 times as price sensitive as food-service buyers, and average processor margins are 33 percent. Controlling for income, education, and population, a 10-percentage-point higher county wholesale-channel share is associated with 16 percent more baseline procurement and 11 percent more total surplus. Under a 20 percent raw-poultry input-cost shock, the same contrast is associated with a 6 percent greater procurement decline and a 4 percent greater buyer-surplus loss, without a detectable additional price response. A merger of the third- and fourth-largest processors raises prices by 6.9 percent. The network thus links local competition and downstream demand organization to market scale and shock exposure.

A Text-Based Measure of Farm Bill Uncertainty and Its Effects on Agricultural Credit

Abstract: This paper develops a text-based measure of Farm Bill uncertainty and studies its effects on U.S. agricultural credit supply. A two-stage BERT classifier applied to newspaper articles yields a Farm Bill Uncertainty Index (FBUI) that isolates language reflecting legislative delay, disagreement, and procedural impasse. The index tracks negotiation cycles, spiking during conflict and declining after enactment. Merging the FBUI with institution-level data for commercial banks, credit unions, and Farm Credit System institutions shows that higher uncertainty reduces agricultural lending, most clearly for commercial banks. Exposure-weighted specifications show reductions in loan amounts and average loan sizes in more exposed markets, while stacked difference-in-differences estimates around the 2008, 2014, and 2018 Farm Bills show faster lending growth after uncertainty declines.

Climate-Conditioned Comparative Advantage, Persistence, and U.S. Corn-Soy Land Allocation (with Xiaodong Du and Xueying Sun)

Abstract: This paper studies how climate-conditioned comparative advantage translates into county-level corn-soy land allocation in the U.S. Corn Belt when production patterns are persistent. Using rain-fed potential-yield simulations from FAO Global Agro-Ecological Zones, it constructs a Relative Input Premium Advantage (RIPA) that compares the potential-yield gains from higher-input management for corn and soybeans. RIPA is associated mainly with persistent cross-county differences in allocation, while year-to-year acreage changes are more closely related to expected relative revenues. A discrete-choice land-allocation model incorporating expected portfolio revenue, RIPA, revenue risk, and prior allocation shows strong state dependence, and counterfactual acreage responses increase substantially when the estimated persistence penalties are removed.

Last Update: October 2026

REFERENCES

Xiaodong (Sheldon) Du (*Ph.D. advisor*)

Professor

Department of Agricultural and Applied Economics

University of Wisconsin–Madison

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David A. Hennessy

Cargill Professor in Economic Systems

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Iowa State University

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